

DEM 601 Introduction to Demographic Analysis

Exercise Set 1

TERM
Fall 2026

SET IN
lecture 1

WORTH
2% of the final grade

DUE
29 September 2026, 23:59

Readings for this lecture

Rowland, Donald T. (2003) *Demographic Methods and Concepts*
Introduction

Hinde, Andrew (1998) *Demographic Methods*
Some Demographic Fundamentals

Newell, Colin (1988) *Methods and Models in Demography*
Introduction · Data Collection

Wachter, Kenneth W. (2014) *Essential Demographic Methods*
Introduction: Why Study Demography?

Carmichael, Gordon A. (2016) *Fundamentals of Demographic Analysis: Concepts, Measures and Methods*
Basic Sources, Concepts, Definitions and Types of Measures

How to hand this in. By 29 September 2026, 23:59. Work in a spreadsheet, R or by hand, as you prefer, but show the arithmetic that got you to each answer, not only the answer. Late work receives no credit, so hand in what you have.

1. The two branches, and stocks and flows

12 marks

- Name the two branches of demography, say what separates them, and say which of the two this course teaches.
- Explain the difference between a **stock** and a **flow**, give one example of each from the balancing equation, and say which of the two a census gives you directly and which a vital registration system gives you.

2. What each source buys, and what it costs

22 marks

For each of the four sources of demographic data, give **one** thing it does better than the others and **one** real limitation.

Source	One strength	One limitation
Census		
Vital registration		
Sample survey		
Big data		

Then answer in two or three sentences: for which of the four can you put a number on the error, and why does that matter more than it first appears?

3. Finding the figures yourself

16 marks

Using the portals named in lecture 1, and listed again below, find:

- the most recent world population;
- the projected world population in 2100.

TurkStat

Türkiye: population, births, deaths, life tables, ABPRS
data.tuik.gov.tr

UN World Population Prospects

Every country, 1950 to 2100, estimates and projections
population.un.org/wpp

The DHS Program

Survey microdata for 90+ countries, fertility and child health
dhsprogram.com

IPUMS International

Harmonised census microdata, free for research
international.ipums.org

Human Mortality Database

Life tables for countries with complete registration
mortality.org

HUIPS

The TDHS series and this institute's own surveys
hips.hacettepe.edu.tr

Human Fertility Database

Period and cohort fertility for countries with complete birth registration
www.humanfertility.org

Eurostat

Europe: population, fertility, mortality and migration, harmonised
ec.europa.eu/eurostat

World Bank Open Data

Population and development indicators for every country
data.worldbank.org

For each figure, **state the source and the year it refers to**. Then compare both with the figures quoted in the lecture, 8.3 billion and 10.2 billion. If your numbers differ, say why. A different number is not necessarily a wrong one.

Every figure needs a source and a year. This is the habit the whole course is graded on. A number without a provenance cannot be checked, and a number that cannot be checked is not evidence.

4. What changes the size of a population?

16 marks

For a **national** population, decide whether each process adds people, removes people, or changes the size not at all. Tick one column.

Process	Adds	Removes	Changes neither
Fertility			
Mortality			
Immigration			
Emigration			
Marriage			
Divorce			
Adoption			
Ageing			

One of these is worth arguing about. In two or three sentences, explain **adoption**: what does it change, if not the size, and when is it the one case on this list that really does move a national headcount?

5. Where would each of them be counted?

16 marks

It is census night. Say where each person is counted under a **de facto** enumeration and under a **de jure** one, and name which method Türkiye used before 2000 and which it uses now.

On census night	De facto	De jure
A student from Sivas, asleep in an Ankara dormitory during term		
A lorry driver from Konya, at a motorway service area		
A family from Gaziantep, on the second night of a holiday in Antalya		
A worker from Ordu who has lived in Germany for nine years		

6. Reading the balancing equation backwards

18 marks

Türkiye's population was 85,664,944 at the end of 2024 and 86,092,168 at the end of 2025.

- How many people did the population gain over that year?
- Look up the registered births and deaths for Türkiye in 2025 on TurkStat. State the figures and where you found them.
- Use the balancing equation to work out what net migration must have been, and say in one sentence what that figure is, given that nobody measured it.

THE POPULATION BLUEPRINT • CHAPTER 1

Across the semester you will build a cumulative demographic profile of one population of your own choosing: a continent, a country or a city. Each chapter uses the methods of the week it is set in, and each is a prerequisite for the next, so they must be done in order.

Türkiye may not be chosen, because every worked example in class uses it.

This first chapter is the choice itself, and it is due with this sheet.

- Name the population you have chosen, and say why it interests you.
- Find out whether the data exists **before** you commit. Name a source that publishes its population count, give the most recent year available, and give the figure.
- Say whether that population has a recent census, a vital registration system, or neither, and how you know.
- Confirm your choice with me by the end of lecture 2. You will use the same population for every chapter, so choose one you can get data for.

Total: 100 marks